

Workshop on Technologies to Support Critical Thinking in an Age of Misinformation

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ABSTRACT

Serious concerns have been raised about social media's and on-line news outlets' contribution to a pandemic of misinformation. The sheer volume and tendency of misinformation to exploit people's cognitive biases have eroded the public's trust in media outlets, governmental institutions, and the democratic process. With Human-Computer Interaction being at the forefront of designing and developing user-facing computing systems, we bear special opportunities to address these issues and work on solutions to mitigate problems arising from misinformation. This workshop brings together designers, developers, and thinkers across disciplines to redefine computing systems by focusing on technologies and applications that instil and nurture critical thinking in their users. By focusing on the problem of misinformation and users' cognitive security, this workshop will sketch out blueprints for systems and interfaces that contribute to advancing technology and media literacy, building critical thinking skills, and helping users telling fake from truth.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Misinformation; Cognitive biases; Human-Computer Interaction

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1 INTRODUCTION

A lot of misinformation is roaming the Internet these days. Characterized as factually incorrect information that is intentionally manipulated to deceive the receiver [11], misinformation often challenges our ability to tell fake from truth. With ties to the war propaganda of the early 20th century, fake news are often accompanied by the fabrication of facts and manipulation of narratives. In comparison to the most extreme and infamous examples of misinformation campaigns conducted by the German Empire during World War 2, the majority of recent misinformation campaigns may not appear to be as harmful. However, today, misinformation can be spread almost instantaneously around the globe, reaching larger audiences than ever before, resulting in unprecedented consequences. Recently, during the COVID-19 pandemic, the message that industrial alcohol kills the COVID-19 virus was circulated over various platforms and resulted in the death of around 480 people, as well as sickness of thousands [8]. New technology has eased the distribution of misinformation, and enabled governments, organisations, and individuals to influence the public opinion or convince the public of a specific agenda, a traditional goal of misinformation campaigns [16]. However, the same technology also offers governments and organisations new avenues to detecting and correcting false information.

1.1 Background

Even though a wide range of tools and approaches to detect and debunk misinformation exists, the answer to why people believe in false information remains complex. Research by Lewandowsky et al. [12] has found that misinformation is often presented as a conspiracy theory making people less likely to accept accurate information from official sources [7]. Similar findings were presented by Van der Linden et al. [19], who exposed participants to both factually correct statements and misinformation and found no change in belief of the participants. The authors concluded that the mere presence of misinformation can cause people to believe in those. Another aspect that needs consideration is a lack of expertise in a specific domain, e.g., science. Layman's knowledge makes it more difficult to make informed decisions, thus non-experts often fall back to simplifying their judgement criteria [3]. Cook et al. [3] uses critical thinking strategies, e.g., method aiming at revealing the deceptive arguments within false claims, to inform consumers of misinformation.

Compared to these, traditional inoculation theory proposes the vaccination of people against persuasive misinformation by building immunity [14]. The irony could not be greater, as the most popular misinformation campaign of our days is grounded in a (now retracted) paper suggesting that the measles, mumps, and rubella (MMR) vaccine potentially causes developmental disorder in children [20]. As in medicine, effective inoculation requires the consumption of weakened misinformation prior to the exposure to the potential misinformation, while requiring the presentation of opposing claims. This latter part bears the risk of causing a backfire effect [15]. In case a person has already been convinced by misinformation, the presentation of the opposing argument often results in a counter-reaction of enforcement of the pre-existing beliefs. As misinformation has long existed in the public sphere, inoculation strategies have to be smart and more targeted.

The reasons for these phenomena can be found in the foundations of human cognition. When encountering new information, we are prone to assessing this new information for its affinity with our existing knowledge [13], which requires cognitive effort. Whereas confirming one's preexisting beliefs creates positive affect, dealing with opposing information triggers a rather negative affective response, as postulated by the *cognitive dissonance* theory [9, 13]. Even if technology achieves to accurately detect and correct misinformation, a plethora of pitfalls inhibits the potential corrective effects. Often, by directly confronting people with misinformation in order to debunk them, facing a confirmation of existing knowledge, no matter if right or wrong, leads to further solidification of the confirmed belief, simply due to the increased amount of evidence at hand [21].

1.2 Motivation

The very same technologies that are used to collect large amounts of personal information and to target users' cognitive vulnerabilities also offer intelligent solutions to the problem of misinformation. Pattern recognition and Natural Language Processing have made fact-checking applications and spam filters more accurate and reliable. Machine Learning, big data, and context-aware computing systems can be used to detect misinformation in-situ, and provide cognitive security Andrade and Yoo [1]. Today, these self-learning systems protect the user and prevent misinformation from finding fertile ground.

Researchers and practitioners in Human-Computer Interaction (HCI) are at the forefront of designing and developing user-facing computing systems. Consequently, we bear special responsibility for working on solutions to mitigate problems arising from misinformation and bias-enforcing interfaces.

This workshop brings together designers, developers, and thinkers across disciplines to redefine computing systems by focusing on technologies, such as recommender and social computing systems. The workshop aims at fostering the development of applications that detect and limit the spread of misinformation as well as interfaces that instill and nurture critical thinking in their users. By focusing on the problem of misinformation and users' cognitive security from a computing and HCI perspective, this workshop will sketch out blueprints for systems and interfaces that contribute

to advancing media literacy, fostering critical thinking skills, and helping users telling fact from fake.

2 THEMES

The following themes are meant to guide authors, contributors, and participants of the workshop:

- **Social Computing System Interventions:** how can targeted systems, learning algorithms, interfaces, and agents be used to prevent, stop, or mitigate misinformation from spreading?
- **Technologies to Foster Critical Thinking:** where market incentives and governmental oversight fail us, how can we re-think technologies to build and reinforce critical thinking skills?
- **Engaging Users:** how can we trigger a design shift from shallow user engagement towards interfaces that prioritise information gain over attention capture and instil better information processing habits in their users?
- **Application Scenarios:** what are application scenarios besides political fact-checking, and social media oversight? How can HCI research contribute to the deployment of ethical, fair, and unbiased systems?
- **Case Studies:** presentation of concrete cases where the impact of user interfaces on users' critical thinking has been investigated, or where interfaces have been evaluated for their effectiveness in mitigating the effects of misinformation on systems, societies, people, media, and computing systems.
- **Tools and Methods:** which methods and tools can help support human cognitive efforts, and promote unbiased and fair contention of pre-existing beliefs?
- **Inoculation Techniques:** the design and creation of systems and methods to help users build resistance against misinformation. This can include the identification and mitigation of fake news and populist messaging, but also ways to build empathy and critical thinking abilities.

3 ORGANIZERS

Tilman Dingler is a lecturer at the School of Computing and Information Systems at the University of Melbourne. He builds and investigates cognition-aware systems and technologies that support users' information processing capabilities. Tilman is initiator of the WAHM series, *i.e.*, a four-workshop series on *Ubiquitous Technologies for Augmenting the Human Mind* [2, 4, 5, 10].

Benjamin Tag is a postdoctoral researcher and associate lecturer at the School of Computing and Information Systems at the University of Melbourne. His research focuses on the conceptualisation of digital emotion regulation, and investigation of human cognition using biometric sensors and psychological test methods. Benjamin a main organizer of the workshop series on *Eyewear Computing* [17, 18].

Philipp Lorenz-Spreen is a network scientist, working at the Center for Adaptive Rationality, who is interested in how and why information is spreading on social media platforms. His current

research turns to the question of how the individual choice environments can be changed such that they translate into positive collective effects of quality information distribution.

Andrew W. Vargo is a research assistant professor at Osaka Prefecture University. His research focuses on the study and creation of collaborative platforms for supporting learning and self-awareness.

Simon Knight is a senior lecturer at the University of Technology Sydney, Faculty of Transdisciplinary Innovation, where he leads the Transformative Learning research theme, and is Director of the Centre for Research on Learning in a Technological Society. His research investigates the role of technology in learning, particularly focusing on critical thinking and dialogue.

Stephan Lewandowsky is a cognitive scientist at the University of Bristol. His main research interest relates to how people respond to misinformation and its correction, and why some people reject well-established scientific knowledge.

4 PRE-WORKSHOP PLANS

This workshop is a specialized follow-up from last year's *Workshop on Detection and Design for Cognitive Biases in People and Computing Systems* [6], which could not take place in person due to the pandemic. We did, however, successfully run a virtual version. This year, we will narrow down the topic range to address the specific problems arising from misinformation.

The goal of the workshop is to foster a research community around the investigation of computing systems and interfaces that support critical thinking and cognitive security, in a world of increasingly effective and targeted distribution of misinformation. As a field in the intersection of Computer Science and Psychology, HCI is well suited for contributing in a variety of ways. We are aiming at building a community with a research agenda around detecting, mitigating, and designing for an increased media-literacy of the general public and to develop new systems that contribute to advancing critical thinking skills, and helping users telling fake from truth.

Prior to the workshop, we will issue a call for participation along with a request for submission of position papers. These will be in the new single-column CHI-format and should include no more than 8 pages, including references. Submissions will be peer-reviewed by a select program committee and curated by the organizing committee with the goal to offer a diverse discussion-fostering program. We plan to use easychair for managing submissions. The workshop will be announced via its website¹ where we will collate materials relevant to the workshop themes a priori. We will additionally disseminate a Call for Participation via the well-known mailing lists and through the organizers' networks.

The organizers will document the results of the discussions and will make the collected information available to workshop participants and the broader community by means of online repositories and a public website. Outcomes include a detailed research agenda forward and a cross-discipline community of international esteem.

5 WORKSHOP STRUCTURE

We propose a one-day workshop with presentation sessions in the morning, development of scenarios in the early afternoon, and group discussions on fundamental challenges in the late afternoon.

5.1 Participants

We plan for 20-30 participants to be part of the workshop (excluding organizers). The workshop's program committee will make a selection of presenters based on submitted position papers and with an eye on inter-disciplinary viewpoints.

5.2 Keynote

The workshop will be kicked off by a prominent researcher or Industry practitioner in the field of misinformation or conspiracy theories.

5.3 Presentations

The organizers will publicly call for participation, and all submissions will be peer-reviewed. Accepted submissions are invited for oral presentation at the workshop. A particular focus will be put on extended discussion times, and the chance for participants to demo technologies and tools to foster engagement and the potential for future collaborations.

5.4 Demonstration Session

We encourage participants to demonstrate research prototypes related to their submissions. Depending on the number of demonstrations we will hold a separate session or demo tools and systems in the coffee breaks.

5.5 Action Groups

Each group (approximately 5-6 members each) will receive a set of tasks that consist of

- analyzing a specific case of misinformation, its spread, and people's susceptibility to it,
- identifying the computing systems involved
- discussing and conceptualizing solutions that can be utilized in this context, and
- developing a conceptual prototype as a solution proposal.

One half of the action groups will work on how to tackle the spread of misinformation (system side), and the other half will work on how to support the information consumer (user side).

5.6 Group Discussion

After the Action Group analysis, we organize a discussion round on identified challenges and opportunities. The organizers will actively engage with the audience to stimulate discussions. We will summarize the key experiences from the workshop and will plan follow up activities.

5.7 Virtual Workshop

Since CHI'21 has been announced to be run virtually, we also intend to run the workshop fully online. We will use a combination of

¹<http://critical-media.org/chi21/>

ZOOM² and Miro³ to keep our participants engaged in different exercises, group-sessions, and feedback-rounds.

6 POST WORKSHOP PLANS

The organizers will document the results of the action groups and discussions and will make the collected information available to workshop participants and the broader community by means of online repositories and a public website. As this workshop is a direct follow-up to the 2020 CHI Workshop on Detection and Design for Cognitive Biases in People and Computing Systems [6], the organizers will summarize the findings and contributions of both workshops. This summary will serve as the foundation for a research agenda on "New Interfaces to Foster the Development of Media Literacy and Critical Thinking of an Informed Public." This research agenda will be submitted to the ACM Interactions Magazine for wider dissemination.

7 CALL FOR PARTICIPATION

Serious concerns have been raised about social media's and online news outlets' contributions to a pandemic of misinformation. The sheer volume, its spread, and the tendency of misinformation to exploit people's cognitive biases have eroded the public's trust in media outlets, governmental institutions, and the democratic process.

The "Workshop on Technologies to Support Critical Thinking in an Age of Misinformation" aims to bring together designers, developers, and thinkers across disciplines to redefine computing systems by focusing on technologies, such as recommender and social computing systems. The workshop aims at fostering the development of applications that detect and limit the spread of misinformation as well as interfaces that instil and nurture critical thinking in their users. By focusing on the problem of misinformation and users' cognitive security from a computing and HCI perspective, this workshop will sketch out blueprints for systems and interfaces that contribute to advancing media literacy, building critical thinking skills, and helping users tell fact from fake.

The following themes are meant to guide authors, contributors, and participants of the workshop:

- **Social Computing System Interventions:** how can targeted systems, learning algorithms, interfaces, and agents be used to prevent, stop, or mitigate misinformation from spreading? How can these systems mitigate adversarial behaviour?
- **Technologies to Foster Critical Thinking:** where market incentives and governmental oversight fail us, how can we re-think technologies to build and reinforce critical thinking skills?
- **Engaging Users:** how can we trigger a design shift from shallow user engagement towards interfaces that prioritise information gain over attention capture and instil better information processing habits in their users?
- **Application Scenarios:** what are application scenarios besides political fact-checking, and social media oversight?

How can HCI research contribute to the deployment of ethical, fair, and unbiased systems?

- **Case Studies:** presentation of concrete cases where the impact of user interfaces on users' critical thinking has been investigated, or where interfaces have been evaluated for their effectiveness in mitigating the effects of misinformation on systems, societies, people, media, and computing systems.
- **Tools and Methods:** which methods and tools can help support human cognitive efforts, and promote unbiased and fair contention of pre-existing beliefs?
- **Inoculation Techniques:** the design and creation of systems and methods to help users build resistance against misinformation. This can include the identification and mitigation of fake news and populist messaging, but also ways to build empathy and critical thinking abilities.

With Human-Computer Interaction being at the forefront of designing and developing user-facing computing systems, we bear special opportunities to address these issues and work on solutions to mitigate problems arising from misinformation. We invite position papers that are concerned with the above-mentioned themes and discuss the role of computing systems, algorithms, and dark pattern designs in user interfaces as potential amplifiers of misinformation. We welcome submissions covering a wide field of topics related to system and user interface design that address issues of misinformation with the goal of mitigating its negative impacts. Submissions are expected to be in the single-column CHI-format and should be no longer than 8 pages, including references. The workshop comprises a mini Paper Committee in charge of peer-reviews.

Further workshop details and the submission link can be found on the workshop's website: <http://critical-media.org/chi21/>.

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